

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q15-0491

Taber 345 kV
July 03, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0491. The Interconnection Customer proposes a 212.8 MW solar photovoltaic facility interconnecting at the Taber 345 kV substation in the Maryland area. The request was received on June 25, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 284 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$7,547,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: JEROME-SANDY SPRING 138 KV
Contingency: TABER-DIXLETA 345 KV
Loading: 228.8% (Rating: 244 MVA)
DFAX: 5.9%
Pre-project loading: 223.7%
2. Facility: MCCALL SUBSTTAION-PEBBLEDLE 138 KV
Contingency: TABER-DIXLETA 345 KV
Loading: 179.5% (Rating: 352 MVA)
DFAX: 7.7%
Pre-project loading: 174.9%
3. Facility: GLEN BURNIE-SANDARIO SUBSTATION 138 KV L
Contingency: DIXLETA-TABER 345 KV
Loading: 172.9% (Rating: 243 MVA)
DFAX: 5.0%
Pre-project loading: 168.5%
4. Facility: TAZEWELL-PORT ROBSON 138 KV line
Contingency: DIXLETA SUBSTATION-TABER 345 KV line
Loading: 169.7% (Rating: 438 MVA)
DFAX: 11.6%
Pre-project loading: 164.1%
5. Facility: MCCALL SUBSTTAION-STILLWATER HYDRO ELECTRIC PROJECT 138 KV
Contingency: TABER-DIXLETA 345 KV
Loading: 162.6% (Rating: 224 MVA)
DFAX: 5.2%
Pre-project loading: 157.7%

6. Facility: TAZEWELL-TABER 138 KV
Contingency: TABER-DIXLETA SUBSTATION 345 KV
Loading: 156.7% (Rating: 520 MVA)
DFAX: 11.6%
Pre-project loading: 152.0%
7. Facility: HIGHGROVE-CTR 138 KV
Contingency: DIXLETA-TABER 345 KV line
Loading: 153.1% (Rating: 417 MVA)
DFAX: 7.2%
Pre-project loading: 149.4%
8. Facility: TABER-PEBBLEDALE 138 KV
Contingency: TABER SUBSTATION-DIXLETA 345 KV
Loading: 150.2% (Rating: 450 MVA)
DFAX: 7.7%
Pre-project loading: 146.6%
9. Facility: STILLWATER HYDRO ELECTRIC PROJECT-SANDARIO 138 KV Line
Contingency: TABER SUBSTATION-DIXLETA 345 KV
Loading: 149.7% (Rating: 232 MVA)
DFAX: 5.2%
Pre-project loading: 144.9%
10. Facility: MARCU HOOK ENERGY LP-IRVINE 500 KV
Contingency: MARCUS HOOK ENERGY LP-GUILBEAU ELECTRIC 500 KV Line
Loading: 122.2% (Rating: 2,306 MVA)
DFAX: 5.9%
Pre-project loading: 121.6%
11. Facility: PORT ROBSON-PAJUILL TC 138 KV
Contingency: DIXLETA-TABER 345 KV L
Loading: 119.9% (Rating: 272 MVA)
DFAX: 8.5%
Pre-project loading: 113.3%
12. Facility: WESTPORT-WHEELABRATOR BALTIMORE REFUSE 138 KV
Contingency: TABER-DIXLETA 345 KV
Loading: 111.7% (Rating: 882 MVA)
DFAX: 8.8%
Pre-project loading: 109.6%
13. Facility: MONTPELIER-DRUM 138 KV
Contingency: TABER-DIXLETA 345 KV
Loading: 110.1% (Rating: 745 MVA)
DFAX: 6.7%
Pre-project loading: 108.2%
14. Facility: TABER-DIXLETA SUBSTATION 345 KV line
Contingency: TABER 345 kV/138 kV XFMR
Loading: 109.5% (Rating: 3,216 MVA)
DFAX: 52.5%
Pre-project loading: 106.0%
15. Facility: HUGHSON-DIXLETA 345 KV Line
Contingency: RUSTIC-DIXLETA 345 KV
Loading: 109.4% (Rating: 2,802 MVA)
DFAX: 34.5%
Pre-project loading: 106.8%
16. Facility: DIXLETA-RUSTIC 345 KV
Contingency: HUGHSON-DIXLETA 345 KV

Loading: 108.5% (Rating: 2,631 MVA)
 DFAX: 30.7%
 Pre-project loading: 106.1%

17. Facility: NORTH POLE-HIGH RIDGE 138 KV line
 Contingency: TABER-DIXLETA 345 KV
 Loading: 105.3% (Rating: 1,303 MVA)
 DFAX: 8.0%
 Pre-project loading: 104.0%

18. Facility: PAJUIL TC-SAN LORENZO 3302 138 KV Line
 Contingency: DIXLETA-TABER 345 KV line
 Loading: 105.1% (Rating: 212 MVA)
 DFAX: 5.7%
 Pre-project loading: 99.3%

19. Facility: JEROME-HIGH RIDGE 138 KV
 Contingency: DIXLETA SUBSTATION-TABER 345 KV Line
 Loading: 104.7% (Rating: 1,346 MVA)
 DFAX: 8.0%
 Pre-project loading: 103.4%

20. Facility: TABER SUBSTATION 345/138kv XFMR
 Contingency: TABER-DIXLETA 345 KV L
 Loading: 101.8% (Rating: 1,392 MVA)
 DFAX: 22.8%
 Pre-project loading: 98.4%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
JEROME-SANDY SPRING 138 KV	Rebuild line with higher-capacity conductor	\$1,264,000	6% (\$69,000)
MCCALL SUBSTTAION-PEBBLEDLE 138 KV	Rebuild line with higher-capacity conductor	\$1,823,000	26% (\$483,000)
GLEN BURNIE-SANDARIO SUBSTATION 138 KV L	Rebuild line with higher-capacity conductor	\$2,046,000	26% (\$542,000)
PORT ROBSON-TAZEWELL 138 KV	Rebuild line with higher-capacity conductor	\$2,029,000	11% (\$225,000)
MCCALL SUBSTTAION-STILLWATER HYDRO ELECTRIC PROJECT 138 KV	Rebuild line with higher-capacity conductor	\$2,834,000	26% (\$751,000)
TAZEWELL-TABER 138 KV	Rebuild line with higher-capacity conductor	\$2,885,000	11% (\$320,000)
HIGHGROVE-CTR 138 KV	Rebuild line with higher-capacity conductor	\$3,497,000	4% (\$155,000)
TABER-PEBBLEDALE 138 KV	Rebuild line with higher-capacity conductor	\$2,727,000	26% (\$723,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SANDARIO-STILLWATER HYDRO ELECTRIC PROJECT SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$1,975,000	26% (\$524,000)
MARCUS HOOK ENERGY LP SUBSTATION-IRVINE 500 KV line	Rebuild line with higher-capacity conductor	\$12,053,000	2% (\$223,000)
PORT ROBSON-PAJUIL TC 138 KV	Rebuild line with higher-capacity conductor	\$2,352,000	27% (\$625,000)
WESTPORT-WHEELABRATOR BALTIMORE REFUSE 138 KV	Rebuild line with higher-capacity conductor	\$600,000	3% (\$17,000)
MONTPELIER-DRUM 138 KV	Rebuild line with higher-capacity conductor	\$10,330,000	3% (\$305,000)
TABER-DIXLETA 345 KV	Reconductor line	\$6,613,000	4% (\$286,000)
HUGHSON-DIXLETA 345 KV Line	Reconductor line	\$11,505,000	3% (\$303,000)
DIXLETA-RUSTIC 345 KV	Reconductor line	\$37,318,000	2% (\$942,000)
HIGH RIDGE-N POLE SUBSTATION 138 KV Line	Reconductor line	\$1,270,000	2% (\$21,000)
SAN LORENZO 3302-PAJUIL TC 138 KV	Reconductor line	\$636,000	27% (\$169,000)
JEROME-HIGH RIDGE 138 KV	Reconductor line	\$582,000	2% (\$9,000)
TABER 345/138kv XFMR	Replace transformer with higher-rated unit	\$9,437,000	9% (\$855,000)
	Total allocated to project		\$7,547,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.